Overall System Architecture

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BEFORE THE FIRST STEP · once per task

Issue + repo
text query; repo at base_commit

Repository index
files · classes · functions · imports (Python ast)

L BM25 lexical

S import structure

RRF
k = 60

Localization hint
top-20 source files + 3 related tests

DURING THE RUN · mini-SWE-agent loop (unchanged)

mini-SWE-agent
LLM decides → bash action → observation → history

Final patch
git diff, one per run

Official SWE-bench harness
FAIL_TO_PASS + PASS_TO_PASS; evaluation only, never shown to the agent

evidence increment

failed command

RRF(L,S,F)
rerank

Failure analysis
traceback → index entities → ranking F

EXPERIMENTAL CONDITIONS

- **A** no hint (original agent)
- **B** hint from L only
- **C** hint from RRF(L, S)
- **D** C + failure-driven reranking